UNIVERSIDAD AUTONOMA DE NUEVO LEON

Facultad de Ciencias Físico Matemáticas

(FCFM)

Unidad de Aprendizaje:

Desarrollo web – Tecnologías Servidor

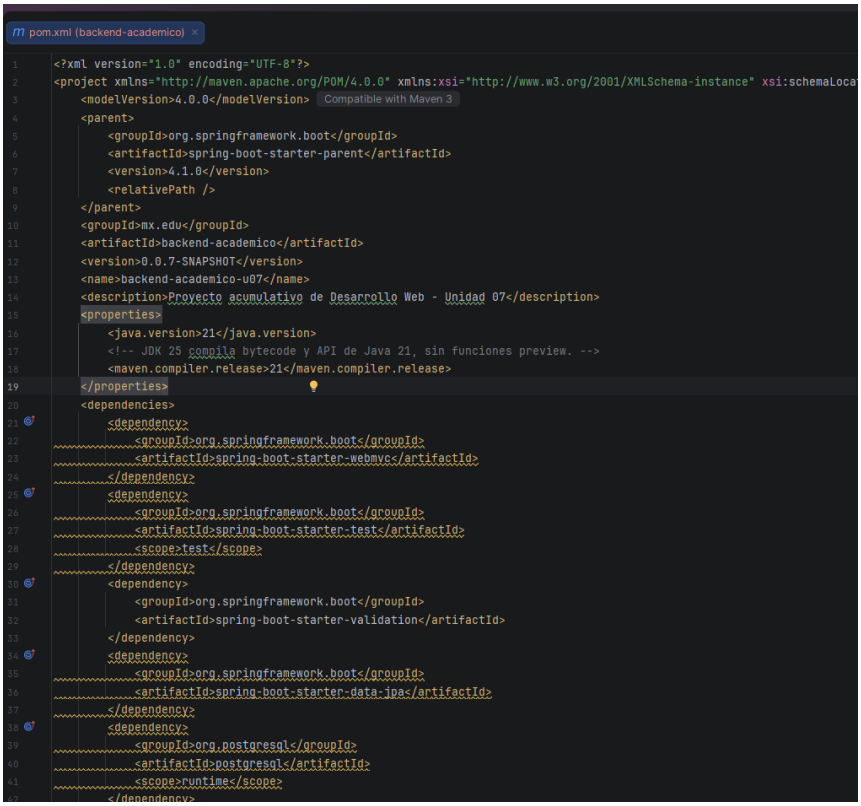
Unidad 07

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Carrera:	LCC

Monterrey N.L. a; 06 de noviembre de 2026

Paso 1 — Construcción y dependencias

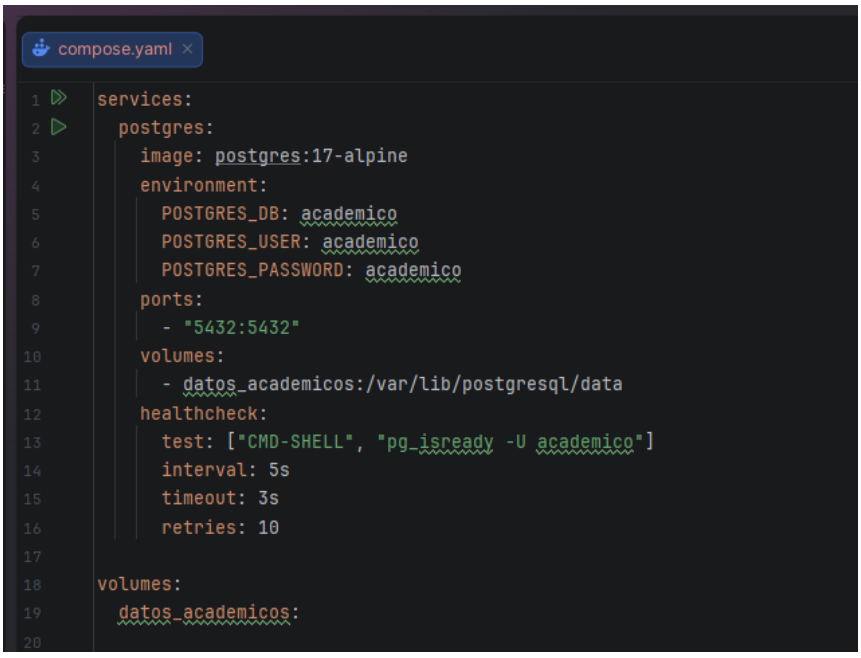
Acción: modificar pom.xml.



```xml
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLoca
    <modelVersion>4.0.0</modelVersion> Compatible with Maven 3
    <parent>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-parent</artifactId>
        <version>4.1.0</version>
        <relativePath />
    </parent>
    <groupId>mx.edu</groupId>
    <artifactId>backend-academico</artifactId>
    <version>0.0.7-SNAPSHOT</version>
    <name>backend-academico-u07</name>
    <description>Proyecto acumulativo de Desarrollo Web - Unidad 07</description>
    <properties>
        <java.version>21</java.version>
        <!-- JDK 25 compila bytecode y API de Java 21, sin funciones preview. -->
        <maven.compiler.release>21</maven.compiler.release>
    </properties>
    <dependencies>
        <dependency>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-starter-webmvc</artifactId>
        </dependency>
        <dependency>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-starter-test</artifactId>
            <scope>test</scope>
        </dependency>
        <dependency>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-starter-validation</artifactId>
        </dependency>
        <dependency>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-starter-data-jpa</artifactId>
        </dependency>
        <dependency>
            <groupId>org.postgresql</groupId>
            <artifactId>postgresql</artifactId>
            <scope>runtime</scope>
        </dependency>
```

Paso 2 — PostgreSQL y red de laboratorio

Acción: crear compose.yaml.



```yaml
services:
  postgres:
    image: postgres:17-alpine
    environment:
      POSTGRES_DB: academico
      POSTGRES_USER: academico
      POSTGRES_PASSWORD: academico
    ports:
      - "5432:5432"
    volumes:
      - datos_academicos:/var/lib/postgresql/data
    healthcheck:
      test: ["CMD-SHELL", "pg_isready -U academico"]
      interval: 5s
      timeout: 3s
      retries: 10

volumes:
  datos_academicos:
```

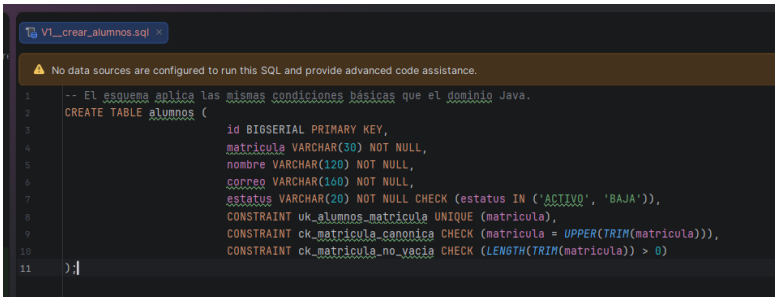
Paso 3 — Valores externos del ambiente

Acción: modificar src/main/resources/application.properties.



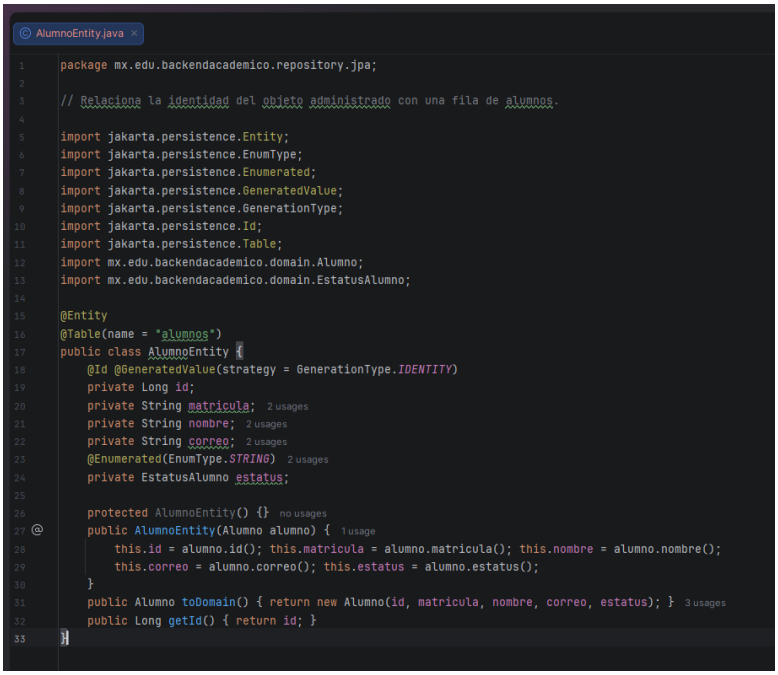
Paso 4 — Esquema inicial de alumnos

Acción: crear src/main/resources/db/migration/V1__crear_alumnos.sql.



Paso 5 — Mapeo de la tabla alumnos

Acción: crear src/main/java/mx/edu/backendacademico/repository/jpa/AlumnoEntity.java.



Paso 6 — Almacén local y sección crítica

Acción: modificar src/main/java/mx/edu/backendacademico/repository/AlumnoRepositoryEnMemoria.java.

AlumnoRepositoryEnMemoria.java

```java
package mx.edu.backendacademico.repository;

import java.util.LinkedHashMap;
import java.util.List;
import java.util.Map;
import java.util.Optional;
import mx.edu.backendacademico.domain.Alumno;
import mx.edu.backendacademico.exception.MatriculaDuplicadaException;
import org.springframework.stereotype.Repository;
import org.springframework.context.annotation.Profile;

@Repository
@Profile("!jpa")
public class AlumnoRepositoryEnMemoria implements AlumnoRepository {
    private final Map<Long, Alumno> datos = new LinkedHashMap<>(); 6 usages
    private long secuencia; 1 usage

    @Override 2 usages
    public synchronized Alumno guardar(Alumno alumno) {
        // 1. Un mismo monitor protege comprobación y escritura como una operación.
        boolean duplicado = datos.values().stream().anyMatch( Alumno actual ->
                actual.matricula().equals(alumno.matricula()) &&
                        !actual.id().equals(alumno.id()));
        if (duplicado) throw new MatriculaDuplicadaException(alumno.matricula());
        // 2. El incremento comparte el monitor; no necesita un AtomicLong adicional.
        long id = alumno.id() == null ? ++secuencia : alumno.id();
        var guardado = new Alumno(id, alumno.matricula(), alumno.nombre(),
                alumno.correo(), alumno.estatus());
        datos.put(id, guardado);
        return guardado;
    }

    @Override 1 usage
    public synchronized List<Alumno> buscarTodos() {
        // Copia estable: no se expone la colección mutable que vive en el bean.
        return List.copyOf(datos.values());
    }

    @Override 1 usage
    public synchronized Optional<Alumno> buscarPorId(Long id) {
        return Optional.ofNullable(datos.get(id));
    }
```

```java
    @Override 1 usage
    public synchronized List<Alumno> buscarTodos() {
        // Copia estable: no se expone la colección mutable que vive en el bean.
        return List.copyOf(datos.values());
    }

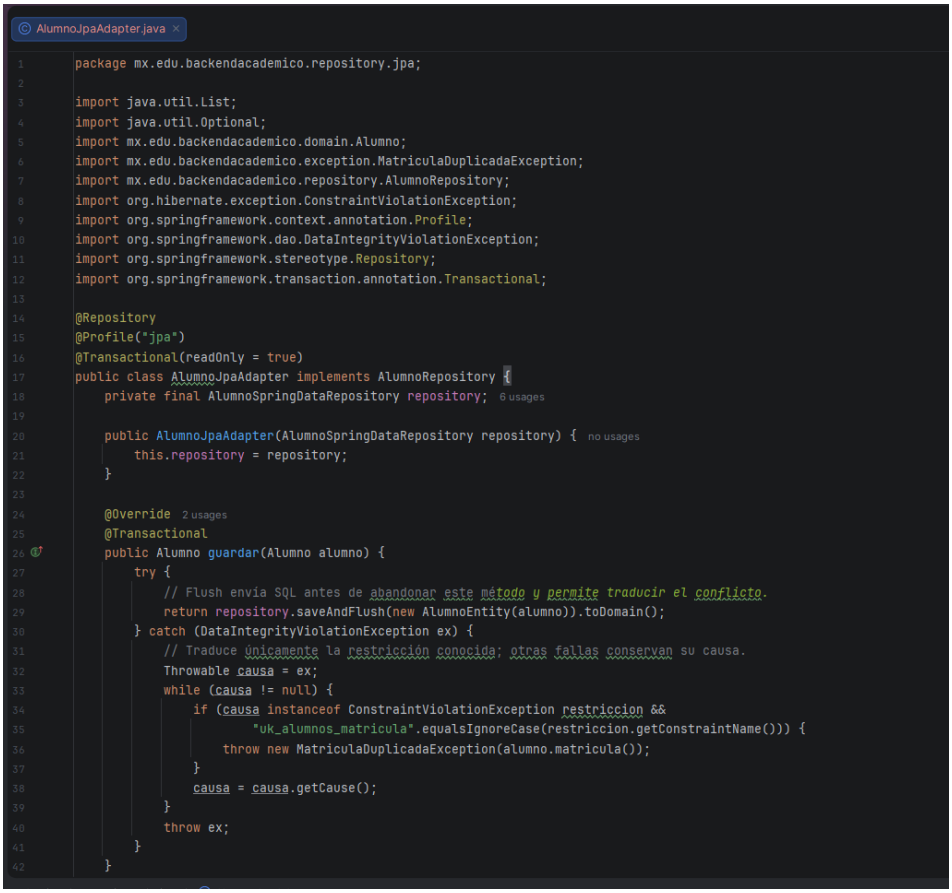
    @Override 1 usage
    public synchronized Optional<Alumno> buscarPorId(Long id) {
        return Optional.ofNullable(datos.get(id));
    }

    @Override 1 usage
    public synchronized boolean existePorMatricula(String matricula) {
        return datos.values().stream().anyMatch( Alumno a -> a.matricula().equalsIgnoreCase(matricula.strip()));
    }

    @Override 1 usage
    public synchronized void eliminarPorId(Long id) {
        datos.remove(id);
    }
}
```

Paso 7 — Sustitución de memoria por JPA

Acción: crear src/main/java/mx/edu/backendacademico/repository/jpa/AlumnoJpaAdapter.java.



```java
package mx.edu.backendacademico.repository.jpa;

import java.util.List;
import java.util.Optional;
import mx.edu.backendacademico.domain.Alumno;
import mx.edu.backendacademico.exception.MatriculaDuplicadaException;
import mx.edu.backendacademico.repository.AlumnoRepository;
import org.hibernate.exception.ConstraintViolationException;
import org.springframework.context.annotation.Profile;
import org.springframework.dao.DataIntegrityViolationException;
import org.springframework.stereotype.Repository;
import org.springframework.transaction.annotation.Transactional;

@Repository
@Profile("jpa")
@Transactional(readOnly = true)
public class AlumnoJpaAdapter implements AlumnoRepository {
    private final AlumnoSpringDataRepository repository;  6 usages

    public AlumnoJpaAdapter(AlumnoSpringDataRepository repository) {  no usages
        this.repository = repository;
    }

    @Override  2 usages
    @Transactional
    public Alumno guardar(Alumno alumno) {
        try {
            // Flush envía SQL antes de abandonar este método y permite traducir el conflicto.
            return repository.saveAndFlush(new AlumnoEntity(alumno)).toDomain();
        } catch (DataIntegrityViolationException ex) {
            // Traduce únicamente la restricción conocida; otras fallas conservan su causa.
            Throwable causa = ex;
            while (causa != null) {
                if (causa instanceof ConstraintViolationException restriccion &&
                        "uk_alumnos_matricula".equalsIgnoreCase(restriccion.getConstraintName())) {
                    throw new MatriculaDuplicadaException(alumno.matricula());
                }
                causa = causa.getCause();
            }
            throw ex;
        }
    }
```



```java
    @Override  1 usage
    public List<Alumno> buscarTodos() {
        // Este listado didáctico se limita a conjuntos pequeños de alumnos.
        return repository.findAll().stream().map(AlumnoEntity::toDomain).toList();
    }

    @Override  1 usage
    public Optional<Alumno> buscarPorId(Long id) {
        return repository.findById(id).map(AlumnoEntity::toDomain);
    }

    @Override  1 usage
    public boolean existePorMatricula(String matricula) {
        return repository.existsByMatriculaIgnoreCase(matricula.strip());
    }

    @Override  1 usage
    @Transactional
    public void eliminarPorId(Long id) {
        repository.deleteById(id);
    }
}
```

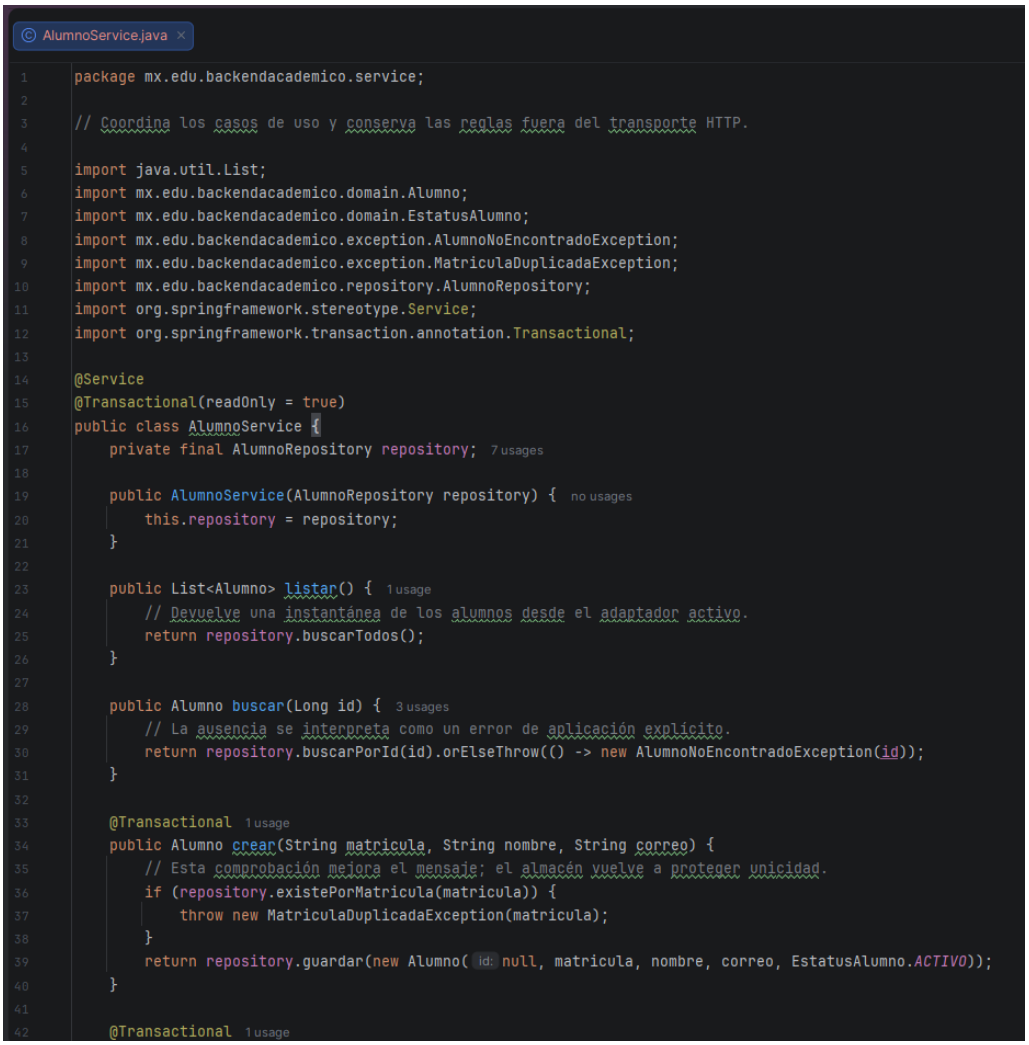
Paso 8 — Repositorio generado por Spring Data

Acción: crear src/main/java/mx/edu/backendacademico/repository/jpa/AlumnoSpringDataRepository.java.



Paso 9 — Casos de uso de alumnos

Acción: modificar src/main/java/mx/edu/backendacademico/service/AlumnoService.java.



Paso 10 — Comprobación del ensamblado

Acción: modificar src/test/java/mx/edu/backendacademico/BackendAcademicoApplicationTests.java.

BackendAcademicoApplicationTests.java

```java
package mx.edu.backendacademico;

import org.junit.jupiter.api.Test;
import org.springframework.boot.test.context.SpringBootTest;

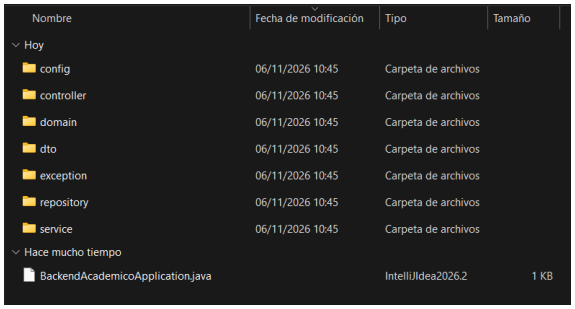
@SpringBootTest(properties = {
        "spring.datasource.url=jdbc:h2:mem:contexto;MODE=PostgreSQL;DB_CLOSE_DELAY=-1",
        "spring.datasource.username=sa", "spring.datasource.password=",
        "spring.flyway.enabled=false", "spring.jpa.hibernate.ddl-auto=create-drop"
})
class BackendAcademicoApplicationTests {
    // Comprueba ensamblado y mapeo JPA. PostgreSQL y migraciones se prueban en U09.
    @Test void cargaElContextoConBaseEfimera() { }
}
```

Cierre de la construcción guiada

Después de comparar, la construcción guiada sí coincide con la carpeta de solución previamente revisada.

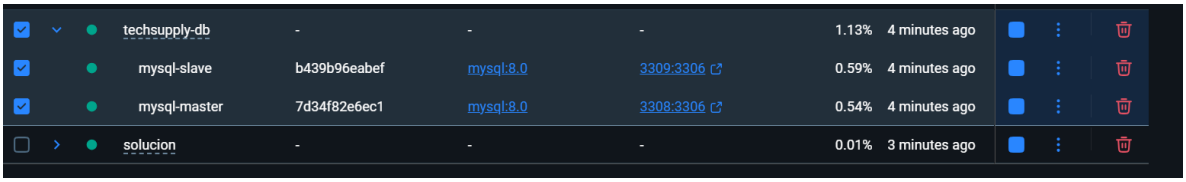


Nombre	Fecha de modificación	Tipo	Tamaño
Hoy			
service	06/11/2026 11:05	Carpeta de archivos	
repository	06/11/2026 11:00	Carpeta de archivos	
config	06/11/2026 10:45	Carpeta de archivos	
controller	06/11/2026 10:45	Carpeta de archivos	
domain	06/11/2026 10:45	Carpeta de archivos	
dto	06/11/2026 10:45	Carpeta de archivos	
exception	06/11/2026 10:45	Carpeta de archivos	
Hace mucho tiempo			
BackendAcademicoApplication.java		IntelliJIdea2026.2	1 KB



Nombre	Fecha de modificación	Tipo	Tamaño
Hoy			
config	06/11/2026 10:45	Carpeta de archivos	
controller	06/11/2026 10:45	Carpeta de archivos	
domain	06/11/2026 10:45	Carpeta de archivos	
dto	06/11/2026 10:45	Carpeta de archivos	
exception	06/11/2026 10:45	Carpeta de archivos	
repository	06/11/2026 10:45	Carpeta de archivos	
service	06/11/2026 10:45	Carpeta de archivos	
Hace mucho tiempo			
BackendAcademicoApplication.java		IntelliJIdea2026.2	1 KB

Comprobación del resultado



techsupply-db	-	-	-	1.13%	4 minutes ago		
mysql-slave	b439b96eabef	mysql:8.0	3309:3306	0.59%	4 minutes ago		
mysql-master	7d34f82e6ec1	mysql:8.0	3308:3306	0.54%	4 minutes ago		
solucion	-	-	-	0.01%	3 minutes ago		

```
PS C:\Users\andre\Downloads\Desarrollo-web-tecnolog-as-servidor\Unidades\Unidad 7\solucion> docker compose up -d
[+] Running 1/1
 ✓ Container solucion-postgres-1  Running

PS C:\Users\andre\Downloads\Desarrollo-web-tecnolog-as-servidor\Unidades\Unidad 7\solucion> docker compose ps
NAME                  IMAGE                COMMAND                  SERVICE    CREATED         STATUS                   PORTS
solucion-postgres-1   postgres:17-alpine   "docker-entrypoint.s…"   postgres   4 minutes ago   Up 4 minutes (healthy)   0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp

PS C:\Users\andre\Downloads\Desarrollo-web-tecnolog-as-servidor\Unidades\Unidad 7\solucion>
```

```
PS C:\Users\andre\Downloads\Desarrollo-web-tecnolog-as-servidor\Unidades\Unidad 7\solucion>
.\mvnw.cmd spring-boot:run
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by
com.google.inject.internal.aop.HiddenClassDefiner (file:/C:/Users/andre/.m2/wrapper/dists/apache-
maven-3.9.11/03d7e36a140982eea48e22c1dcac01d8862b2550b2939e09a0809bbc5182a5bc/lib/guice-5.1.0-
classes.jar)
WARNING: Please consider reporting this to the maintainers of class
com.google.inject.internal.aop.HiddenClassDefiner
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
[INFO] Scanning for projects...
[INFO]
[INFO] ----------------------< mx.edu:backend-academico >----------------------
[INFO] Building backend-academico-u07 0.0.7-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] >>> spring-boot:4.1.0:run (default-cli) > test-compile @ backend-academico >>>
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ backend-academico ---
[INFO] Copying 1 resource from src\main\resources to target\classes
[INFO] Copying 1 resource from src\main\resources to target\classes
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ backend-academico ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ backend-academico ---
[INFO] skip non existing resourceDirectory C:\Users\andre\Downloads\Desarrollo-web-tecnolog-as-
servidor\Unidades\Unidad 7\solucion\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ backend-academico ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] <<< spring-boot:4.1.0:run (default-cli) < test-compile @ backend-academico <<<
[INFO]
[INFO]
[INFO] --- spring-boot:4.1.0:run (default-cli) @ backend-academico ---
[INFO] Attaching agents: []

  .   ____          _            __ _ _
 /\\ / ___'_ __ _ _(_)_ __  __ _ \ \ \ \
( ( )\___ | '_ | '_| | '_ \/ _` | \ \ \ \
 \\/  ___)| |_)| | | | | || (_| |  ) ) ) )
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 :: Spring Boot ::                (v4.1.0)

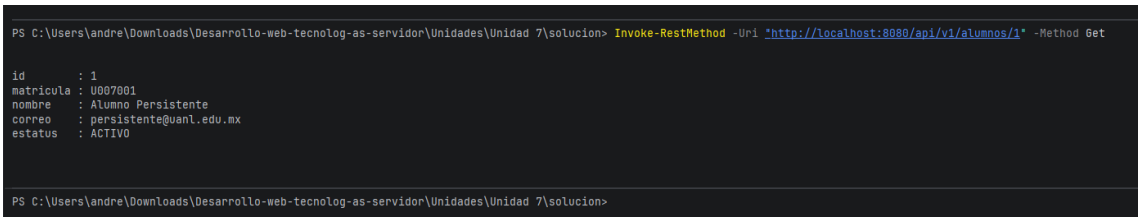
2026-11-06T23:37:28.365-06:00  INFO 16852 --- [backend-academico] [           main]
m.e.b.BackendAcademicoApplication        : Starting BackendAcademicoApplication using Java 24.0.2
with PID 16852 (C:\Users\andre\Downloads\Desarrollo-web-tecnolog-as-servidor\Unidades\Unidad
7\solucion\target\classes started by andre in C:\Users\andre\Downloads\Desarrollo-web-tecnolog-as-
servidor\Unidades\Unidad 7\solucion)
2026-11-06T23:37:28.368-06:00  INFO 16852 --- [backend-academico] [           main]
m.e.b.BackendAcademicoApplication        : No active profile set, falling back to 1 default
profile: "jpa"
2026-11-06T23:37:29.040-06:00  INFO 16852 --- [backend-academico] [           main]
```

```
.s.d.r.c.RepositoryConfigurationDelegate : Bootstrapping Spring Data JPA repositories in DEFAULT
mode.
2026-11-06T23:37:29.097-06:00  INFO 16852 --- [backend-academico] [           main]
.s.d.r.c.RepositoryConfigurationDelegate : Finished Spring Data repository scanning in 48 ms.
Found 1 JPA repository interface.
2026-11-06T23:37:29.515-06:00  INFO 16852 --- [backend-academico] [           main]
o.s.boot.tomcat.TomcatWebServer          : Tomcat initialized with port 8080 (http)
2026-11-06T23:37:29.531-06:00  INFO 16852 --- [backend-academico] [           main]
o.apache.catalina.core.StandardService   : Starting service [Tomcat]
2026-11-06T23:37:29.532-06:00  INFO 16852 --- [backend-academico] [           main]
o.apache.catalina.core.StandardEngine    : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-11-06T23:37:29.597-06:00  INFO 16852 --- [backend-academico] [           main]
b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in
1167 ms
2026-11-06T23:37:29.852-06:00  INFO 16852 --- [backend-academico] [           main]
com.zaxxer.hikari.HikariDataSource       : HikariPool-1 - Starting...
2026-11-06T23:37:30.005-06:00  INFO 16852 --- [backend-academico] [           main]
com.zaxxer.hikari.pool.HikariPool        : HikariPool-1 - Added connection
org.postgresql.jdbc.PgConnection@66ec4409
2026-11-06T23:37:30.007-06:00  INFO 16852 --- [backend-academico] [           main]
com.zaxxer.hikari.HikariDataSource       : HikariPool-1 - Start completed.
2026-11-06T23:37:30.034-06:00  INFO 16852 --- [backend-academico] [           main]
org.flywaydb.core.FlywayExecutor         : Database: jdbc:postgresql://localhost:5432/academico
(PostgreSQL 17.11)
2026-11-06T23:37:30.120-06:00  INFO 16852 --- [backend-academico] [           main]
o.f.core.internal.command.DbValidate     : Successfully validated 1 migration (execution time
00:00.026s)
2026-11-06T23:37:30.142-06:00  INFO 16852 --- [backend-academico] [           main]
o.f.core.internal.command.DbMigrate      : Current version of schema "public": 1
2026-11-06T23:37:30.145-06:00  INFO 16852 --- [backend-academico] [           main]
o.f.core.internal.command.DbMigrate      : Schema "public" is up to date. No migration necessary.
2026-11-06T23:37:30.249-06:00  INFO 16852 --- [backend-academico] [           main]
org.hibernate.orm.jpa                    : HHH008540: Processing PersistenceUnitInfo [name:
default]
2026-11-06T23:37:30.311-06:00  INFO 16852 --- [backend-academico] [           main]
org.hibernate.orm.core                   : HHH000001: Hibernate ORM core version 7.4.1.Final
2026-11-06T23:37:30.772-06:00  INFO 16852 --- [backend-academico] [           main]
o.s.o.j.p.SpringPersistenceUnitInfo      : No LoadTimeWeaver setup: ignoring JPA class transformer
2026-11-06T23:37:30.866-06:00  INFO 16852 --- [backend-academico] [           main]
org.hibernate.orm.connections.pooling    : HHH10001005: Database info:
        Database JDBC URL [jdbc:postgresql://localhost:5432/academico]
        Database driver: PostgreSQL JDBC Driver
        Database dialect: PostgreSQLDialect
        Database version: 17.11
        Default catalog/schema: academico/public
        Autocommit mode: undefined/unknown
        Isolation level: READ_COMMITTED [default READ_COMMITTED]
        JDBC fetch size: none
        Pool: DataSourceConnectionProvider
        Minimum pool size: undefined/unknown
        Maximum pool size: undefined/unknown
2026-11-06T23:37:31.713-06:00  INFO 16852 --- [backend-academico] [           main]
org.hibernate.orm.core                   : HHH000489: No JTA platform available (set
'hibernate.transaction.jta.platform' to enable JTA platform integration)
2026-11-06T23:37:31.767-06:00  INFO 16852 --- [backend-academico] [           main]
j.LocalContainerEntityManagerFactoryBean : Initialized JPA EntityManagerFactory for persistence
unit 'default'
2026-11-06T23:37:31.786-06:00  INFO 16852 --- [backend-academico] [           main]
m.e.backendacademico.config.CicloDeVida  : Bean inicializado en 2026-11-07T05:37:31.786927300Z
2026-11-06T23:37:31.849-06:00  INFO 16852 --- [backend-academico] [           main]
o.s.d.j.r.query.QueryEnhancerFactories   : Hibernate is in classpath; If applicable, HQL parser
will be used.
2026-11-06T23:37:32.386-06:00  INFO 16852 --- [backend-academico] [           main]
o.s.boot.tomcat.TomcatWebServer          : Tomcat started on port 8080 (http) with context path
'/'
2026-11-06T23:37:32.390-06:00  INFO 16852 --- [backend-academico] [           main]
m.e.b.BackendAcademicoApplication        : Started BackendAcademicoApplication in 4.536 seconds
(process running for 5.015)
```

Crear alumno



Consultar



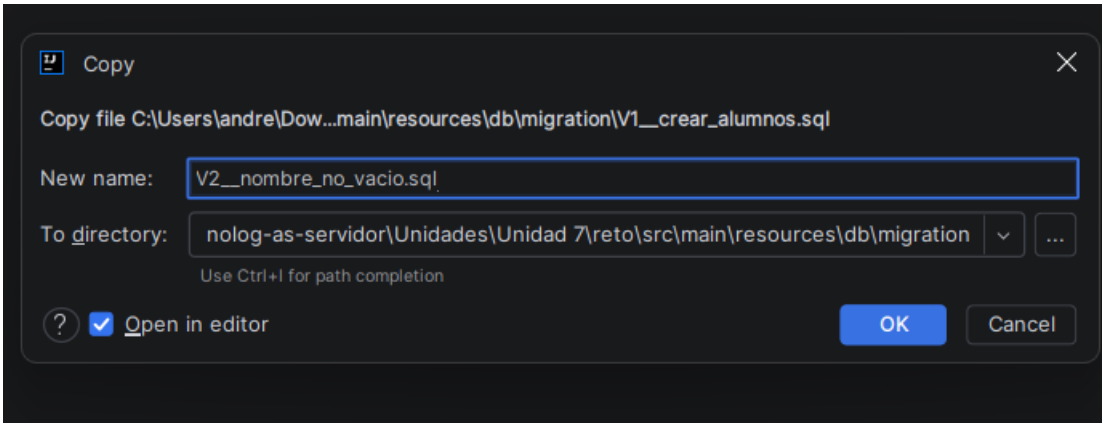
Se reinicio la ejecucion y se hizo la consulta nuevamente

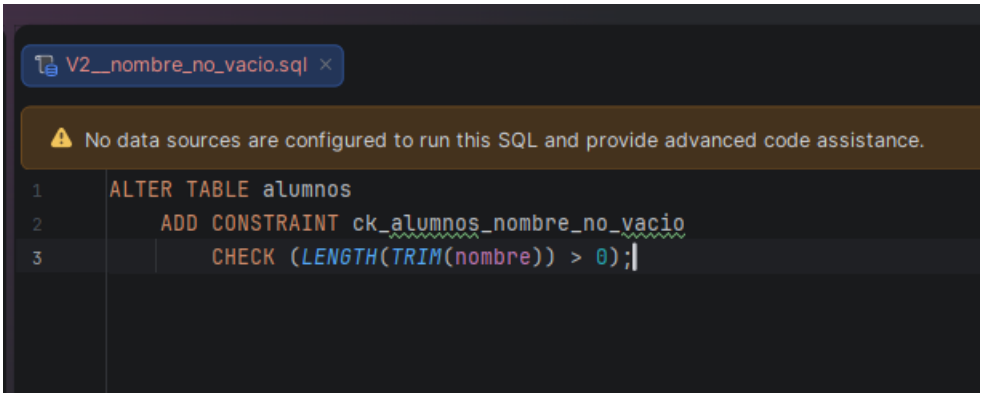


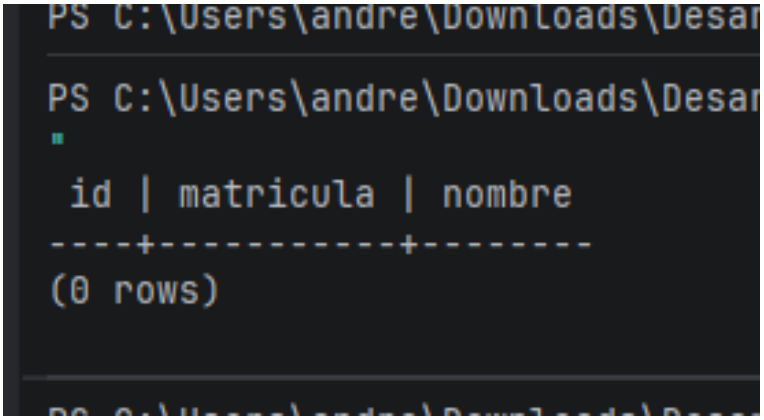
El alumno permanece en el contenedor a pesar del reinicio de springboot



Reto


Copy
Copy file C:\Users\andre\Dow...main\resources\db\migration\V1__crear_alumnos.sql
New name:
V2__nombre_no_vacio.sql
To directory:
nolog-as-servidor\Unidades\Unidad 7\reto\src\main\resources\db\migration
Use Ctrl+I for path completion
Open in editor
OK
Cancel


V2__nombre_no_vacio.sql
No data sources are configured to run this SQL and provide advanced code assistance.
ALTER TABLE alumnos
ADD CONSTRAINT ck_alumnos_nombre_no_vacio
CHECK (LENGTH(TRIM(nombre)) > 0);


PS C:\Users\andre\Downloads\Desar
PS C:\Users\andre\Downloads\Desar
id | matricula | nombre
----+-----------+--------
(0 rows)

```
PS C:\Users\andre\Downloads\Desarrollo-web-tecnolog-as-servidor\Unidades\Unidad 7\reto> .\mvnw.cmd
spring-boot:run
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by
com.google.inject.internal.aop.HiddenClassDefiner (file:/C:/Users/andre/.m2/wrapper/dists/apache-
maven-3.9.11/03d7e36a140982eea48e22c1dcac01d8862b2550b2939e09a0809bbc5182a5bc/lib/guice-5.1.0-
classes.jar)
WARNING: Please consider reporting this to the maintainers of class
com.google.inject.internal.aop.HiddenClassDefiner
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
[INFO] Scanning for projects...
[INFO]
[INFO] ----------------------< mx.edu:backend-academico >----------------------
[INFO] Building backend-academico-u07 0.0.7-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] >>> spring-boot:4.1.0:run (default-cli) > test-compile @ backend-academico >>>
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ backend-academico ---
[INFO] Copying 1 resource from src\main\resources to target\classes
[INFO] Copying 2 resources from src\main\resources to target\classes
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ backend-academico ---
[INFO] Recompiling the module because of added or removed source files.
[INFO] Compiling 21 source files with javac [debug parameters release 21] to target\classes
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ backend-academico ---
[INFO] skip non existing resourceDirectory C:\Users\andre\Downloads\Desarrollo-web-tecnolog-as-
servidor\Unidades\Unidad 7\reto\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ backend-academico ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 2 source files with javac [debug parameters release 21] to target\test-classes
[INFO]
[INFO] <<< spring-boot:4.1.0:run (default-cli) < test-compile @ backend-academico <<<
[INFO]
[INFO]
[INFO] --- spring-boot:4.1.0:run (default-cli) @ backend-academico ---
[INFO] Attaching agents: []

  .   ____          _            __ _ _
 /\\ / ___'_ __ _ _(_)_ __  __ _ \ \ \ \
( ( )\___ | '_ | '_| | '_ \/ _` | \ \ \ \
 \\/  ___)| |_)| | | | | || (_| |  ) ) ) )
  '  |____| .__|_| |_|_| |_\__, | / / / /
 =========|_|==============|___/=/_/_/_/

 :: Spring Boot ::                (v4.1.0)

2026-11-06T23:48:00.340-06:00  INFO 8324 --- [backend-academico] [           main]
m.e.b.BackendAcademicoApplication        : Starting BackendAcademicoApplication using Java 24.0.2
with PID 8324 (C:\Users\andre\Downloads\Desarrollo-web-tecnolog-as-servidor\Unidades\Unidad
7\reto\target\classes started by andre in C:\Users\andre\Downloads\Desarrollo-web-tecnolog-as-
servidor\Unidades\Unidad 7\reto)
2026-11-06T23:48:00.343-06:00  INFO 8324 --- [backend-academico] [           main]
m.e.b.BackendAcademicoApplication        : No active profile set, falling back to 1 default
profile: "jpa"
2026-11-06T23:48:01.087-06:00  INFO 8324 --- [backend-academico] [           main]
.s.d.r.c.RepositoryConfigurationDelegate : Bootstrapping Spring Data JPA repositories in DEFAULT
mode.
2026-11-06T23:48:01.150-06:00  INFO 8324 --- [backend-academico] [           main]
.s.d.r.c.RepositoryConfigurationDelegate : Finished Spring Data repository scanning in 54 ms.
Found 1 JPA repository interface.
2026-11-06T23:48:01.698-06:00  INFO 8324 --- [backend-academico] [           main]
o.s.boot.tomcat.TomcatWebServer          : Tomcat initialized with port 8080 (http)
2026-11-06T23:48:01.717-06:00  INFO 8324 --- [backend-academico] [           main]
o.apache.catalina.core.StandardService   : Starting service [Tomcat]
2026-11-06T23:48:01.718-06:00  INFO 8324 --- [backend-academico] [           main]
o.apache.catalina.core.StandardEngine    : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-11-06T23:48:01.804-06:00  INFO 8324 --- [backend-academico] [           main]
b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in
1407 ms
2026-11-06T23:48:02.103-06:00  INFO 8324 --- [backend-academico] [           main]
com.zaxxer.hikari.HikariDataSource       : HikariPool-1 - Starting...
2026-11-06T23:48:02.286-06:00  INFO 8324 --- [backend-academico] [           main]
```

```
com.zaxxer.hikari.pool.HikariPool        : HikariPool-1 - Added connection
org.postgresql.jdbc.PgConnection@66ec4409
2026-11-06T23:48:02.288-06:00  INFO 8324 --- [backend-academico] [           main]
com.zaxxer.hikari.HikariDataSource       : HikariPool-1 - Start completed.
2026-11-06T23:48:02.320-06:00  INFO 8324 --- [backend-academico] [           main]
org.flywaydb.core.FlywayExecutor         : Database: jdbc:postgresql://localhost:5432/academico
(PostgreSQL 17.11)
2026-11-06T23:48:02.434-06:00  INFO 8324 --- [backend-academico] [           main]
o.f.core.internal.command.DbValidate     : Successfully validated 2 migrations (execution time
00:00.040s)
2026-11-06T23:48:02.459-06:00  INFO 8324 --- [backend-academico] [           main]
o.f.core.internal.command.DbMigrate      : Current version of schema "public": 1
2026-11-06T23:48:02.479-06:00  INFO 8324 --- [backend-academico] [           main]
o.f.core.internal.command.DbMigrate      : Migrating schema "public" to version "2 - nombre no
vacio"
2026-11-06T23:48:02.524-06:00  INFO 8324 --- [backend-academico] [           main]
o.f.core.internal.command.DbMigrate      : Successfully applied 1 migration to schema "public",
now at version v2 (execution time 00:00.023s)
2026-11-06T23:48:02.657-06:00  INFO 8324 --- [backend-academico] [           main]
org.hibernate.orm.jpa                    : HHH008540: Processing PersistenceUnitInfo [name:
default]
2026-11-06T23:48:02.735-06:00  INFO 8324 --- [backend-academico] [           main]
org.hibernate.orm.core                   : HHH000001: Hibernate ORM core version 7.4.1.Final
2026-11-06T23:48:03.234-06:00  INFO 8324 --- [backend-academico] [           main]
o.s.o.j.p.SpringPersistenceUnitInfo      : No LoadTimeWeaver setup: ignoring JPA class transformer
2026-11-06T23:48:03.334-06:00  INFO 8324 --- [backend-academico] [           main]
org.hibernate.orm.connections.pooling    : HHH10001005: Database info:
        Database JDBC URL [jdbc:postgresql://localhost:5432/academico]
        Database driver: PostgreSQL JDBC Driver
        Database dialect: PostgreSQLDialect
        Database version: 17.11
        Default catalog/schema: academico/public
        Autocommit mode: undefined/unknown
        Isolation level: READ_COMMITTED [default READ_COMMITTED]
        JDBC fetch size: none
        Pool: DataSourceConnectionProvider
        Minimum pool size: undefined/unknown
        Maximum pool size: undefined/unknown
2026-11-06T23:48:04.302-06:00  INFO 8324 --- [backend-academico] [           main]
org.hibernate.orm.core                   : HHH000489: No JTA platform available (set
'hibernate.transaction.jta.platform' to enable JTA platform integration)
2026-11-06T23:48:04.370-06:00  INFO 8324 --- [backend-academico] [           main]
j.LocalContainerEntityManagerFactoryBean : Initialized JPA EntityManagerFactory for persistence
unit 'default'
2026-11-06T23:48:04.395-06:00  INFO 8324 --- [backend-academico] [           main]
m.e.backendacademico.config.CicloDeVida  : Bean inicializado en 2026-11-07T05:48:04.395143500Z
2026-11-06T23:48:04.483-06:00  INFO 8324 --- [backend-academico] [           main]
o.s.d.j.r.query.QueryEnhancerFactories   : Hibernate is in classpath; If applicable, HQL parser
will be used.
2026-11-06T23:48:05.177-06:00  INFO 8324 --- [backend-academico] [           main]
o.s.boot.tomcat.TomcatWebServer          : Tomcat started on port 8080 (http) with context path
'/'
2026-11-06T23:48:05.187-06:00  INFO 8324 --- [backend-academico] [           main]
m.e.b.BackendAcademicoApplication        : Started BackendAcademicoApplication in 5.364 seconds
(process running for 5.833)
```

```
PS C:\Users\andre\Downloads\Desarrollo-web-tecnolog-as-servidor\Un
istory ORDER BY installed_rank;"
 installed_rank | version |   description   | success
----------------+---------+-----------------+---------
              1 | 1       | crear alumnos   | t
              2 | 2       | nombre no vacio | t
(2 rows)
```

```
PS C:\Users\andre\Downloads\Desarrollo-web-tecnolog-as-servidor\Unidades\Unidad 7\reto> docker exec solucion-postgres-1 psql -U
 id | matricula |       nombre       |         correo          | estatus
----+-----------+--------------------+-------------------------+---------
  1 | U007001   | Alumno Persistente | persistente@uanl.edu.mx | ACTIVO
(1 row)

PS C:\Users\andre\Downloads\Desarrollo-web-tecnolog-as-servidor\Unidades\Unidad 7\reto>
```

La migración V2 agregó una restricción CHECK para impedir que se registren nombres vacíos o compuestos únicamente por espacios. Se conservó V1 y se registró el nuevo cambio mediante Flyway. Antes de aplicarlo se comprobó que no existieran registros incompatibles. Si los hubiera, PostgreSQL rechazaría la nueva restricción hasta corregirlos. Finalmente, se verificó que el alumno previamente registrado permaneciera en la tabla, demostrando que la estructura se modificó sin eliminar los datos existentes.